

Gravity Books DWH

Data Model Documentation

ITI Egypt — Data Engineering Track

1. OLTP Source Schema — gravity_books

Entity-Relationship Overview

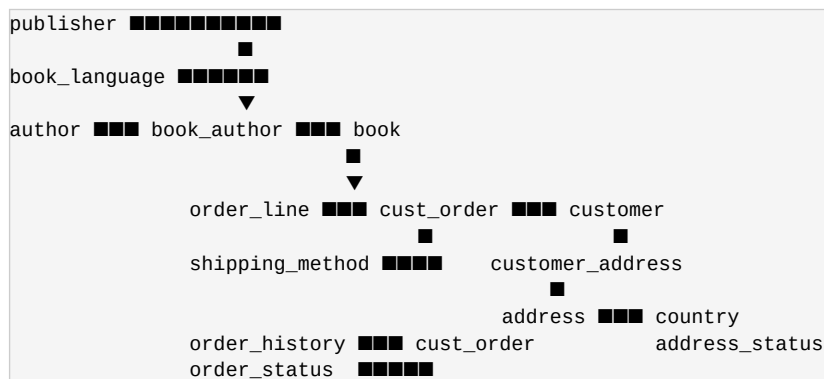


Table Definitions

customer

Column	Type	Notes
customer_id	int PK	
first_name	varchar	
last_name	varchar	
email	varchar	

address

Column	Type	Notes
address_id	int PK	
street_number	varchar	
street_name	varchar	
city	varchar	
country_id	int FK	→ country

customer_address

Column	Type	Notes
customer_id	int FK	→ customer

Column	Type	Notes
address_id	int FK	→ address
status_id	int FK	→ address_status

book

Column	Type	Notes
book_id	int PK	
title	varchar	
isbn13	varchar	
language_id	int FK	→ book_language
num_pages	int	
publication_date	date	
publisher_id	int FK	→ publisher

cust_order

Column	Type	Notes
order_id	int PK	
customer_id	int FK	→ customer
shipping_method_id	int FK	→ shipping_method
order_date	datetime	
dest_address_id	int FK	→ address

order_line

Column	Type	Notes
line_id	int PK	
order_id	int FK	→ cust_order
book_id	int FK	→ book
price	decimal	

order_history

Column	Type	Notes
history_id	int PK	
order_id	int FK	→ cust_order
status_id	int FK	→ order_status
status_date	datetime	Event timestamp

order_status — Lookup Values

status_id	status_value
1	Order Received

Column	Type	Notes
street_number	varchar(10)	
street_name	varchar(200)	
city	varchar(100)	
country_id_bk	int	Original country ID
country_name	varchar(200)	Denormalized
start_date	date	SCD2
end_date	date	SCD2
is_current	tinyint	SCD2

dim_customer_address (Bridge Table)

Column	Type	Notes
customer_SK_FK	int PK (composite)	FK → dim_customer
address_SK_FK	int PK (composite)	FK → dim_address
status_id_bk	int	
address_status	varchar(30)	e.g. Delivery / Current

dim_book (SCD Type 2)

Column	Type	Notes
book_SK	int IDENTITY PK	
book_id_bk	int	Business key
title	varchar(400)	
isbn13	varchar(13)	
language_id_bk	int	
language_code	varchar(8)	
language_name	varchar(50)	Denormalized
num_pages	int	
publication_date	date	
publisher_id_bk	int	
publisher_name	nvarchar(1000)	Denormalized
start_date	date	SCD2
end_date	date	SCD2
is_current	tinyint	SCD2

dim_author

Column	Type	Notes
author_SK	int IDENTITY PK	
author_id_bk	int	

Column	Type	Notes
author_name	varchar(400)	

dim_book_author (Bridge — many-to-many)

Column	Type	Notes
book_SK_FK	int PK (composite)	FK → dim_book
author_SK_FK	int PK (composite)	FK → dim_author

dim_shipping_method (SCD Type 2)

Column	Type	Notes
shipping_method_SK	int IDENTITY PK	
method_id_bk	int	
method_name	varchar(100)	
cost	decimal(6,2)	
start_date	date	SCD2
end_date	date	SCD2
is_current	tinyint	SCD2

Fact Tables

fact_order

Column	Type	Notes
order_fact_SK	int IDENTITY PK	
book_SK_FK	int FK	→ dim_book
customer_SK_FK	int FK	→ dim_customer
shipping_method_SK_FK	int FK	→ dim_shipping_method
address_SK_FK	int FK	→ dim_address
order_id_DD	int	Degenerate dimension
order_line_id_DD	int	Degenerate dimension
order_date_SK_FK	int FK	→ Dim_Date
order_time_SK_FK	int FK	→ Dim_Time
price	decimal(5,2)	Order line price

fact_order_process

Column	Type	Notes
order_id_BK	int PK	Business key (order_id from OLTP)
received_date_SK_FK	int FK	→ Dim_Date
pending_date_SK_FK	int FK	→ Dim_Date
inprogress_date_SK_FK	int FK	→ Dim_Date

Column	Type	Notes
delivered_date_SK_FK	int FK	→ Dim_Date
cancelled_date_SK_FK	int FK	→ Dim_Date
returned_date_SK_FK	int FK	→ Dim_Date
customer_SK_FK	int FK	→ dim_customer
shipping_method_SK_FK	int FK	→ dim_shipping_method
current_status	nvarchar(50)	Latest order status
days_to_ship	int	Days from received to in-progress
days_to_deliver	int	Days from received to delivered

3. ETL Load Order — SSIS Packages

All packages must execute in dependency order to maintain referential integrity:

Step	Package	Target Table	Dependencies
1	Customer_dim_v001.dtsx	dim_customer	None
2	Address_dim_v001.dtsx	dim_address	None
3	Customer_address_dim_v001.dtsx	dim_customer_address	dim_customer, dim_address
4	Book_dim_v001.dtsx	dim_book	None
5	Author_dim_v001.dtsx	dim_author	None
6	Book_author_dim_v001.dtsx	dim_book_author	dim_book, dim_author
7	Shipping_method_dim.dtsx	dim_shipping_method	None
8	Order_fact_v001.dtsx	fact_order	All dims + Dim_Date + Dim_Time
9	Order_process_fact_v003.dtsx	fact_order_process	dim_customer, dim_shipping_method, Dim_Date

Note: Steps 1–7 (dimensions) must all complete before running Steps 8–9 (facts). Steps 1–2 and 4–5 are independent and can run in parallel.

Connection Managers

File	Purpose
LocalHost.gravity_books.conmgr	OLTP source — update server name before deploying
LocalHost.gravity_books_DWH.conmgr	DWH destination — update server name before deploying

4. SSAS Cube — Gravity Books DWH

The SSAS Multidimensional cube is built over **fact_order_process** and provides OLAP-style analysis of order fulfillment performance across the full order lifecycle.

Measures

Measure	Type	Description
days_to_ship	Numeric	Days from order received to shipped (in-progress)
days_to_deliver	Numeric	Days from order received to delivered
Order Count	Count	Number of orders (implicit count measure)

Role-Playing Date Dimensions

The single **Dim_Date** table is reused 5 times, each representing a different point in the order lifecycle:

Role Name	Fact Column	Business Meaning
Dim Date (Order)	order_date_SK	When the order was placed
Dim Date received	received_date_SK_FK	When order entered the system
Dim Date pending	pending_date_SK_FK	When order moved to pending
Dim Date inprogress	inprogress_date_SK_FK	When shipping started
Dim Date cancelled	cancelled_date_SK_FK	When order was cancelled
Dim Date returned	returned_date_SK_FK	When order was returned

Typical Analysis Examples

Analysis	Description
Delivery performance by shipping method	Average days_to_deliver grouped by Dim Shipping Method
Order volume over time	Count of orders by month using Dim Date received
Order funnel / status breakdown	Distribution of current_status across time
Customer order history	Customer orders by year using Dim Date (Order)
Cancellation analysis	Cancelled orders by book category and date